

we have to study from where ai ml standard level

If you mean “AI/ML ko standard industry level tak kaha se padhna chahiye?”, then I’d recommend a structured path rather than randomly doing courses.

AI/ML Standard-Level Roadmap

1. Python + Data Handling

- Python fundamentals
- NumPy
- Pandas
- Matplotlib / Seaborn
- Jupyter Notebook
- Basic statistics

2. Mathematics for ML

- Linear algebra → vectors, matrices, eigenvalues
- Probability → distributions, Bayes theorem
- Statistics → mean, variance, hypothesis testing
- Calculus → derivatives, gradients
- Optimization → gradient descent

3. Classical Machine Learning

Learn **both theory + implementation from scratch**:

- Linear Regression
- Logistic Regression
- KNN
- Naive Bayes
- Decision Trees
- Random Forest
- SVM
- K-Means
- PCA
- Gradient Boosting / XGBoost
- Ensemble methods

For every algorithm, understand:

Intuition → Mathematics → From-scratch implementation → sklearn → Dataset → Evaluation

4. Deep Learning

- Neural Networks
- Forward/backpropagation
- Activation functions
- Loss functions
- Optimizers
- CNN
- RNN/LSTM
- Transformers
- PyTorch

5. Modern AI / GenAI

- Embeddings
- Attention
- Transformers
- LLMs
- Prompt engineering
- RAG
- Vector databases
- Fine-tuning
- Agents
- Evaluation

6. ML Engineering

This is what separates a course-level learner from an industry-level ML engineer:

- Git/GitHub
- APIs with FastAPI
- Docker
- Model serving
- ML pipelines
- Experiment tracking
- Data/version management
- Cloud basics
- MLOps

Where I would study

Area	Recommended source
Python/Data	Kaggle + official docs

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ML fundamentals	Andrew Ng – Machine Learning Specialization
Mathematics	3Blue1Brown + StatQuest
ML algorithms	Hands-On Machine Learning
Deep Learning	fast.ai + PyTorch tutorials
Deep learning theory	Stanford CS231n / CS224n
LLMs	Hugging Face courses + original papers
Practical ML	Kaggle
MLOps	Made With ML

But **don't study all of these simultaneously**.

Since you're already doing **Python, Pandas, NumPy and DSA**, I'd put you roughly at the beginning of the data/ML preparation stage. Your next progression should be:

Python → NumPy/Pandas → Statistics → ML → Deep Learning → GenAI → MLOps → Projects

And importantly, don't spend 6 months only watching lectures. For each major topic, **build something**.

If your goal is **AI/ML placement/internship**, I can also give you a **6–8 month industry-level roadmap with exact courses, YouTube playlists, books, projects, and weekly schedule**.